Comparing Nearest Neighbor, Clarke-Wright Savings, and Harmony Search for the CVRP

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This project used GenAI for generating graphs, debugging code, understanding Python concepts, and Git/GitHub troubleshooting. All algorithm logic, experimental decisions, analysis, and results were produced and verified by the team.

Problem Overview

- Vehicle Routing Problem (VRP): optimize delivery routes for a fleet of vehicles.
- Focus: Capacitated Vehicle Routing Problem (CVRP).
- Vehicles have fixed capacity constraints.
- Goal: minimize total travel distance while satisfying all customer demand.
- CVRP is NP-hard and closely related to the Traveling Salesman Problem.
- Heuristic methods provide practical solutions for large instances.

Algorithms

Nearest Neighbor (NN)

- Greedy baseline heuristic.
- Repeatedly visits the closest feasible unvisited customer.

Clarke-Wright Savings (CW)

- Begins with one route per customer.
- Merges routes using maximum distance savings.

Harmony Search (HS)

- Metaheuristic based on harmony memory.
- Generates and improves candidate solutions through memory consideration, pitch adjustment, and local search.

Research Questions

- How does runtime change as customer count increases?
- Which algorithm provides the best balance between solution quality and speed?
- Does each algorithm's gap from the best-known solution stay consistent as problem size grows?

Expected Outcomes

- NN should be fastest.
- Harmony Search should provide stronger route quality.
- Clarke-Wright may offer the best trade-off.

Experimental Design

- Benchmark dataset: CVRPLIB instances.
- Problem sizes: approximately 30–200 customers.
- Each algorithm tested 10 times per instance.

Metrics:

- Percentage gap from best-known solution.
- Execution runtime (milliseconds).

Goal:

- Compare quality, speed, and scalability.

Theoretical Complexity

Algorithm	Complexity	Main Bottleneck
Nearest Neighbor	$\Theta(n^2)$	Scanning unvisited customers repeatedly
Clarke-Wright Savings	$\Theta(n^2 \log n)$	Sorting savings values
Harmony Search	$\Theta(In^3)$	Iterative local search and solution evaluation

Nearest Neighbor | Experimental Results

data file	num_customers	num_routes	total_distance	best value	gap_percent	avg_runtime_ms	trials
data/A-n32-k5.vrp	31	5	1146.4	784	46.22	0.5312	10
data/E-n51-k5.vrp	50	5	711.5	521	36.56	1.1336	10
data/E-n101-k8.vrp	100	8	1174.36	815	44.09	3.3567	10
data/M-n200-k16.vrp	199	17	2034.4	N/A	N/A	16.4016	10

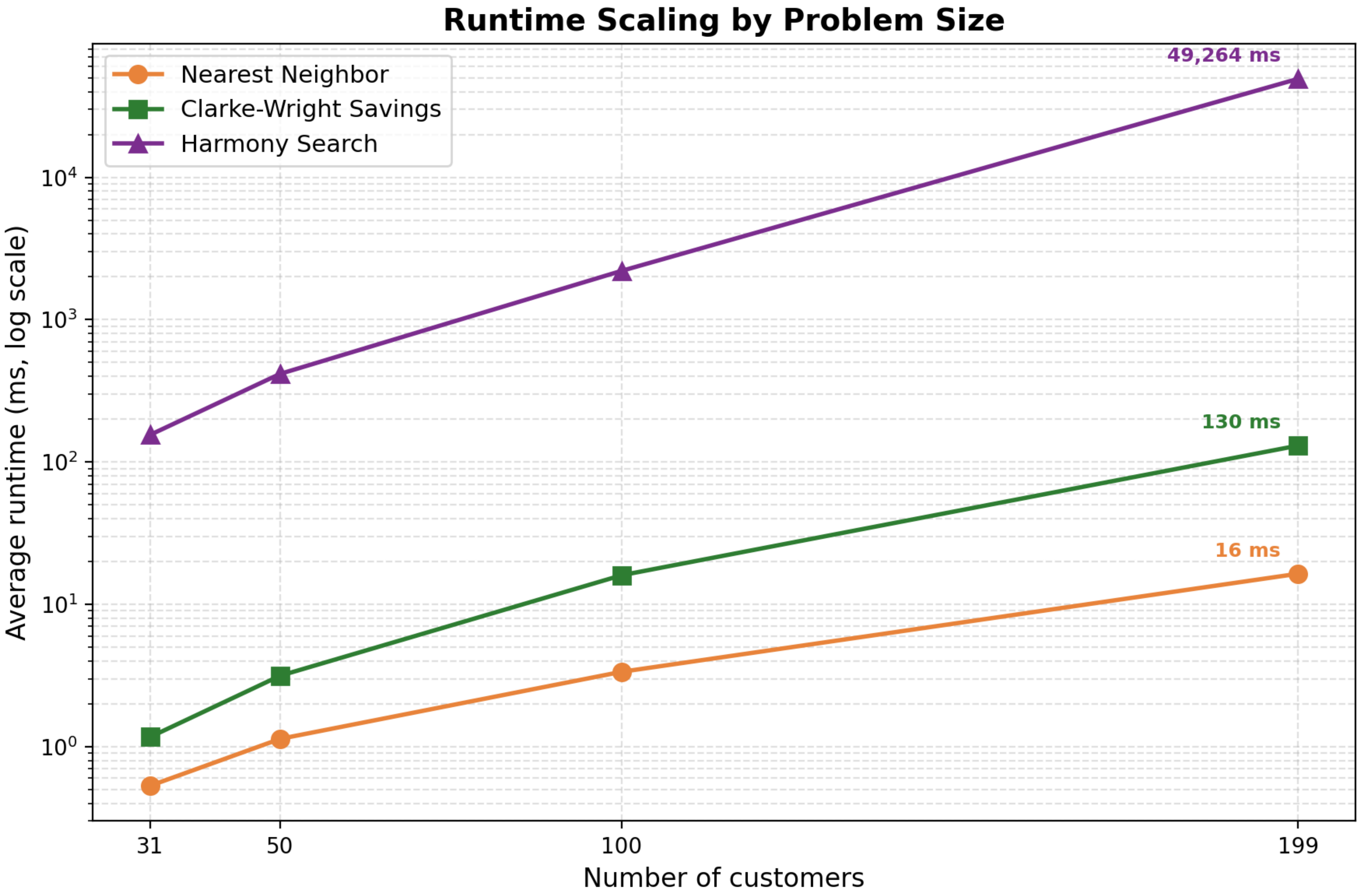
Clarke-Wright Savings

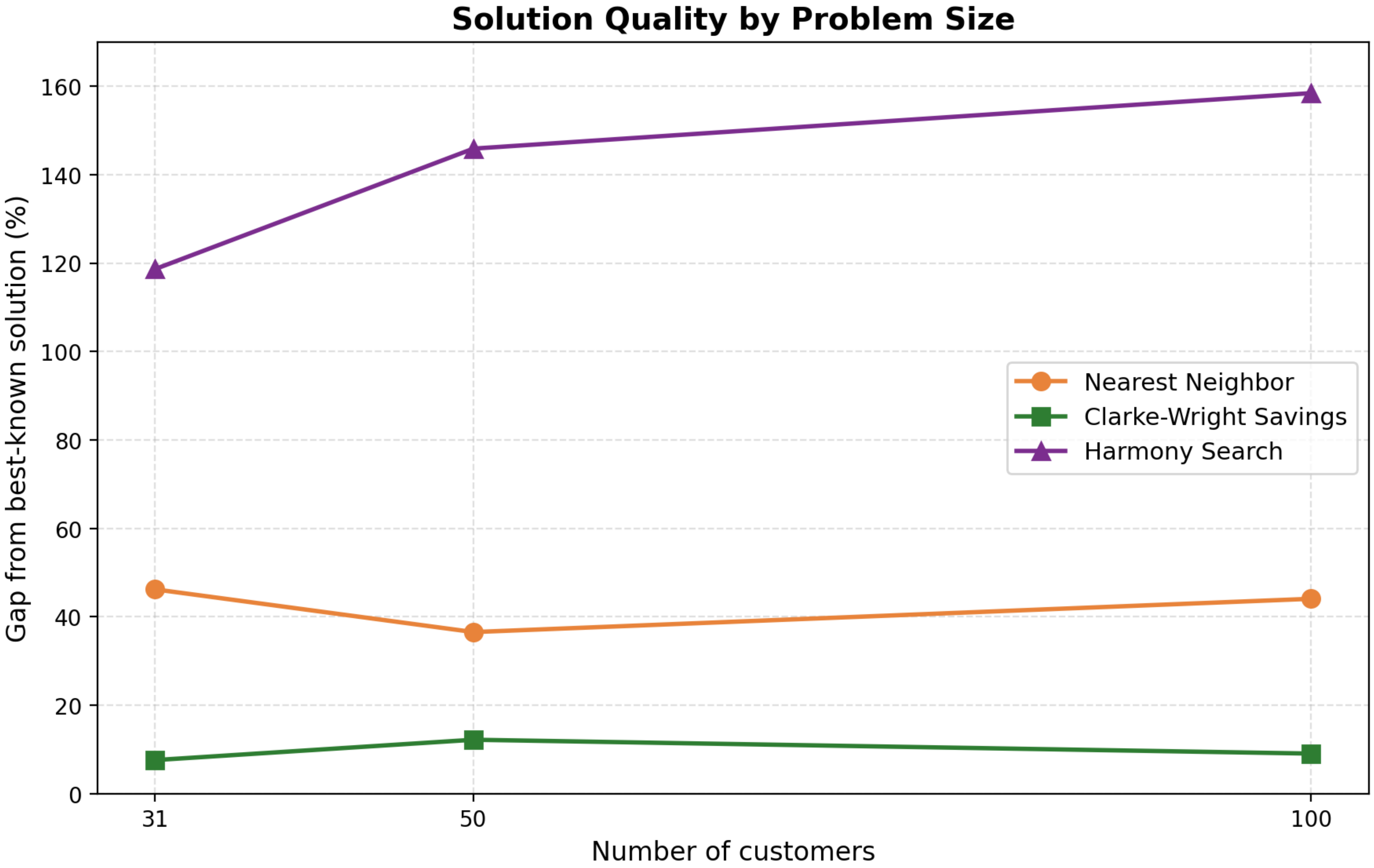
Experimental Results

clarke_wright	data/A-n32-k5.vrp	31	5	843.69	784	7.61	1.1698	10
clarke_wright	data/E-n51-k5.vrp	50	6	584.64	521	12.21	3.1485	10
clarke_wright	data/E-n101-k8.vrp	100	8	889.0	815	9.08	16.0064	10
clarke_wright	data/M-n200-k16.vrp	199	17	1395.74	N/A	N/A	130.4101	10

Harmony Search | Experimental Results

harmony_search	data/A-n32-k5.vrp	31	5	1714.2	784	118.65	155.3155	10
harmony_search	data/E-n51-k5.vrp	50	6	1281.13	521	145.9	416.1114	10
harmony_search	data/E-n101-k8.vrp	100	8	2106.33	815	158.45	2204.2205	10
harmony_search	data/M-n200-k16.vrp	199	17	4915.27	N/A	N/A	49263.7119	10


Runtime Scaling by Problem Size
Nearest Neighbor
Clarke-Wright Savings
Harmony Search
49,264 ms
130 ms
16 ms
Average runtime (ms, log scale)
10^0
10^1
10^2
10^3
10^4
31
50
100
199
Number of customers



M-n200-k16 (199 customers) omitted — no published best-known value.

Analysis & Interpretation

- Observed runtime growth aligns with theoretical complexity.
- NN scales efficiently but sacrifices route quality.
- CW maintains strong solution quality with moderate runtime costs.
- HS incurs substantially higher computational cost.
- Results suggest Harmony Search performance may depend heavily on parameter tuning and implementation details.
- Overall best balance: Clarke-Wright Savings.

Limitations & Future Work

Limitations

- Experiments limited to instances up to 200 customers.
- Deterministic Harmony Search had no randomized selection which limited exploration.
- Wall-clock time is machine-dependent.
- Only four instances tested; three had best-known values.

Future Work

- Evaluate larger CVRPLIB benchmarks.
- Tune Harmony Search parameters.
- Introduce randomized memory consideration and exploration strategies.

Conclusion

YouTube link: [Comparing Nearest Neighbor, Clarke-Wright Savings & Harmony Search to Solve CVRP]

Key Findings

Nearest Neighbor was the fastest algorithm.

Clarke-Wright offered the best quality-to-runtime trade-off.

Harmony Search required the most computation and showed sensitivity to implementation choices.

Takeaway

Classic heuristics can outperform sophisticated metaheuristics when the latter aren't carefully tuned, and algorithm choice matters more than problem size for solution quality.